

Gloria (Pin-Hsun) Lian

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EDUCATION

B. S., PSYCHOLOGY

National Taiwan University, Taipei, Taiwan

Sep.2014 - Jun. 2019

M. S., INSTITUTE OF MEDICAL DEVICE AND IMAGING, COLLEGE OF MEDICINE

National Taiwan University, Taipei, Taiwan

Sep.2022 - Jun. 2024

M. S., DEPARTMENT OF COUNSELING AND CLINICAL PSYCHOLOGY

National Dong Hwa University, Hualien, Taiwan

Sep. 2022 - Sep. 2025

RESEARCH EXPERIENCE

Graduate Research Assistant

Laboratory of AI in Predictive Medicine, National Taiwan University

Jul. 2021-Jul. 2024

Explainable AI for Counterfactual Anomaly Detection in Medical Images

Designed diffusion-model-based deep generative models for adversarial image synthesis to identify symptom-related saliencies in osteoarthritis MRI scans; applied contrastive learning with counterfactual generation to perform unsupervised 3D isotropic enhancement and anomaly detection from longitudinal clinical data, enhancing assessment of degenerative musculoskeletal conditions.

Facial Expression Generation and Action Unit Analysis

Developed a two-stage generative framework combining GANs and diffusion models for high-fidelity facial expression translation with identity preservation; conducted Action Unit analysis to enhance model interpretability, validate emotional targeting, and improve robustness of facial expression generation.

Teaching Assistant, Machine Learning in Biomedical Applications

Institute of Medical Device and Imaging, College of Medicine, NTU

Sep.2023-Feb.2024

Assisted in programming and counseling for a doctoral course on deep learning applications in medical imaging.

Research Assistant

Laboratory of Genetic Statistics, Institute of Biomedical Sciences, Academia Sinica (Taiwan)

Aug.2020-Jun.2021

Participated in phenome-wide association study (PheWAS) related research and the establishment of polygenic risk score (PRS) models to investigate the convoluted interplay between genetics and phenotypes of complex diseases in Taiwanese population. Performing data visualization and statistical analysis of genomic data using the language of R and Python.

Clinical Research Assistant

Drug Prevention and Control Center, Taipei City Hospital (Taiwan)

Nov.2019-Jul.2020

Conducted interviews and psychological assessments to gather data for a survey on substance abuse within the sexual-minority community, specifically investigating the correlation between methamphetamine dependency and participation in chemsex among men who have sex with men.

PUBLICATION

Pin-Hsun Lian, Tzu-Yi Chuang, Yu-Chen Kuo, Gary Han Chang., 2025. Localizing Knee Pain via Explainable Bayesian Generative Models and Counterfactual MRI: Data from the Osteoarthritis Initiative. *Journal of Imaging Informatics in Medicine*. (Conditionally accepted)

Pin-Hsun Lian, Jui-Yo Hsu, Tzu-Yi Chuang, Yi-Hwa Chen, Gary Han Chang., 2025. Self-Supervised Isotropic Resolution Enhancement and Denoising Diffusion for 3D Abnormalities Detection from Anisotropic MRI. *Computerized Medical Imaging and Graphics*. (Conditionally accepted)

CONFERENCE PRESENTATIONS AND AWARDS

Lian, P.-H., T. Chuang, Y.-H. Yen and G. Chang., 2023. Automatic Identification of Pain-Associated Imaging Abnormalities from Knee MRI Imaging by Deep Generative Networks. *Osteoarthritis and Cartilage, Volume 31, S9 - S10*

Concepts and Laboratory Implementation in Psychology: A Case Study on Empathy. Outstanding Research and Creative Award for College Students' Research Project, Ministry of Science and Technology, Taipei, Taiwan, 2019

WORKING EXPERIENCE

Machine Learning Engineer

Brainmate (part-time)

Jan.2025-present

Co-developed an AI-powered cognitive rehabilitation platform for individuals with dementia, implementing object detection models (YOLO and SAM2) with real-time inference to support interactive therapy modules.

Developed deep learning inference pipelines using PyTorch and Docker; collaborated with engineers via GitLab repositories to deploy models to edge devices and GCP cloud services.

Collaborated with clinicians, engineers, and designers, contributed psychological expertise to task design and usability refinement, and integrated LLM APIs to support interactive product features.

Clinical Psychologist Fellowship

Taipei Veterans General Hospital, Taoyuan Branch

Jul.2024-June.2025

Conducted psychological assessments and psychotherapy sessions to aid in diagnosis and treatment planning.

Led psychological groups for both acute and chronic inpatients, facilitating activities designed to improve emotional regulation and social skills.

PROFESSIONAL SKILLS

- AI Techniques: PyTorch, GANs, Diffusion Models, YOLO, SAM2, LLM implementations
- Software Tools: Docker, RESTful API development, Google cloud services
- Programming Language: Python, C++, R
- Language: English (C1 level, TOEFL iBT 103), Mandarin